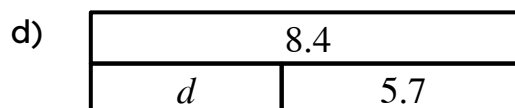
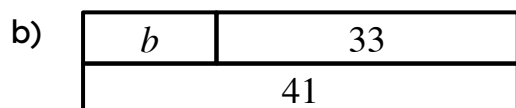
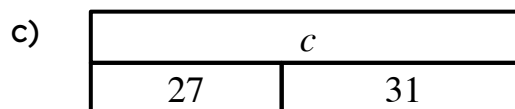
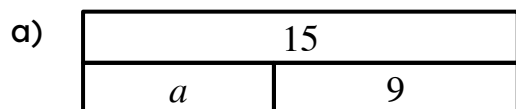




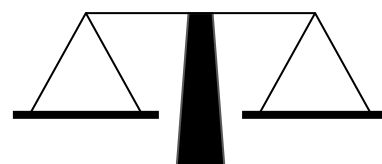
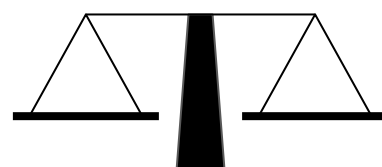
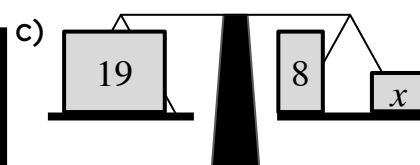
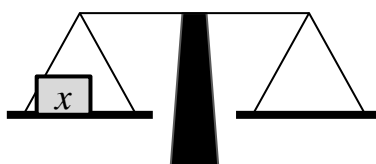
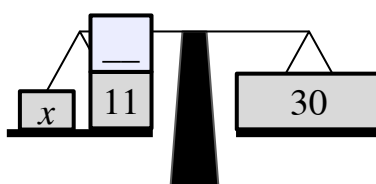
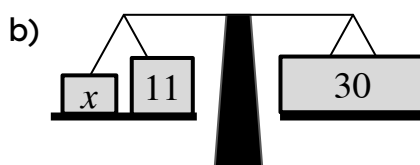
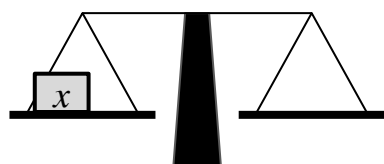
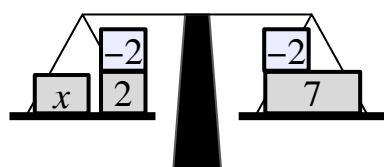
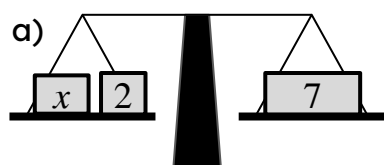
Solving simple linear equations with an additive step

Task A: Additive steps using representations

1) Find the value of the unknown in each bar model. Explain how you know.



2) Complete the balance scales to show the additive step needed to find the solution. What is the solution for each equation?



Task B: Writing solutions algebraically

1) Solve the following equations.

a)

$$x + 1 = 12$$

d)

$$x + 13 = 11$$

b)

$$x + 3.6 = 17.8$$

e)

$$x - 10 = -8.4$$

c)

$$x - 18 = 42$$

f)

$$x + \frac{3}{5} = -\frac{9}{10}$$

2) Spot the mistakes in the following methods to solve these equations.

a)

$$x + 7 = -3$$

$$x + 7 + (7) = -3 + (7)$$

$$x = 4$$

c)

$$x - 5 = 10$$

$$x - 5 + 10 = 10 + 10$$

$$x + 5 = 20$$

b)

$$x + 7.1 = 23.5$$

$$x + 7.1 + (-7.1) = 23.5$$

$$x = 23.5$$

d)

$$x - 23 = 23$$

$$x - 23 + 23 = 23 - 23$$

$$x = 0$$

